星火终审分享

王一璐

1. 写作

Courtesy of Dr. Cindy Koerner

2. 答辩

Courtesy of Dr. Maiwen Caudron-Herger

3. 其他

1. 写作

Courtesy of Dr. Cindy Koerner

2. 答辩

Courtesy of Dr. Maiwen Caudron-Herger

3. 其他

写作顺序 ≠ 文档顺序

文档结构：

1. Title
2. Abstract
3. Introduction
4. Methods
5. Results & Figures/Tables
6. Discussion
7. References

（通常）写作顺序：

1. 初步Title
2. Figures/Tables & Results
3. Methods
4. Introduction
5. Discussion
6. Abstract
7. References
8. 最终Title

各部分的功能

Figures/Tables:

- 讲故事的核心
- 想清楚“说服别人最少需要什么数据”
- 可以加一张总结性的示意图

Introduction:

1. 背景与问题
2. 重要性：复述相关研究，提供 rationale
3. 假设/实验设计概述/关键结果预告

Results:

按段落组织，每段四件事：1. 为什么做这个实验 → 2. 简述实验 → 3. 主要结果 → 4. 小结论

Discussion:

结构灵活，可以包括：总结核心发现，对比相关研究，讨论优势/局限，放入更大背景（理论/应用），给出展望

Methods:

- 足够别人复现
- 顺序与Results一致
- 不写结果

Abstract:

- 通常三件事：Rationale → Methods & Results → Conclusion
- 通常不用缩写，不放参考文献

各部分的功能

Figures/Tables:

- 讲故事的核心
- 想清楚“说服别人最少需要什么数据”
- 可以加一张总结性的示意图

Results:

按段落组织，每段四件事：1. 为什么做这个实验 → 2. 简述实验 → 3. 主要结果 → 4. 小结论

Methods:

- 足够别人复现
- 顺序与Results一致
- 不写结果

Introduction:

1. 背景与问题
2. 重要性：复述相关研究，提供 rationale
3. 假设/实验设计概述/关键结果预告

Discussion:

结构灵活，可以包括：总结核心发现，对比相关研究，讨论优势/局限，放入更大背景（理论/应用），给出展望

Abstract:

- 通常三件事：Rationale → Methods & Results → Conclusion
- 通常不用缩写，不放参考文献

各部分的功能

Figures/Tables:

- 讲故事的核心
- 想清楚“说服别人最少需要什么数据”
- 可以加一张总结性的示意图

Results:

按段落组织，每段四件事：1. 为什么做这个实验 → 2. 简述实验 → 3. 主要结果 → 4. 小结论

Methods:

- 足够别人复现
- 顺序与Results一致
- 不写结果

Introduction:

1. 背景与问题
2. 重要性：复述相关研究，提供 rationale
3. 假设/实验设计概述/关键结果预告

Discussion:

结构灵活，可以包括：总结核心发现，对比相关研究，讨论优势/局限，放入更大背景（理 论/应用），给出展望

Abstract:

- 通常三件事：Rationale → Methods & Results → Conclusion
- 通常不用缩写，不放参考文献

各部分的功能

Figures/Tables:
- 讲故事的核心
- 想清楚“说服别人最少需要什么数据”
- 可以加一张总结性的示意图

Introduction:
1. 背景与问题
2. 重要性：复述相关研究，提供 rationale
3. 假设/实验设计概述/关键结果预告

Results:
按段落组织，每段四件事：1. 为什么做这个实验 → 2. 简述实验 → 3. 主要结果 → 4. 小结论

Discussion:
结构灵活，可以包括：总结核心发现，对比相关研究，讨论优势/局限，放入更大背景（理论/应用），给出展望

Methods:
- 足够别人复现
- 顺序与Results一致
- 不写结果

Abstract:
- 通常三件事：Rationale → Methods & Results → Conclusion
- 通常不用缩写，不放参考文献

各部分的功能

Figures/Tables:

- 讲故事的核心
- 想清楚“说服别人最少需要什么数据”
- 可以加一张总结性的示意图

Results:

按段落组织，每段四件事：1. 为什么做这个实验 → 2. 简述实验 → 3. 主要结果 → 4. 小结论

Methods:

- 足够别人复现
- 顺序与Results一致
- 不写结果

Introduction:

1. 背景与问题
2. 重要性：复述相关研究，提供 rationale
3. 假设/实验设计概述/关键结果预告

Discussion:

结构灵活，可以包括：总结核心发现，对比相关研究，讨论优势/局限，放入更大背景（理论/应用），给出展望

Abstract:

- 通常三件事：Rationale → Methods & Results → Conclusion
- 通常不用缩写，不放参考文献

各部分的功能

Figures/Tables:

- 讲故事的核心
- 想清楚“说服别人最少需要什么数据”
- 可以加一张总结性的示意图

Results:

按段落组织，每段四件事：1. 为什么做这个实验 → 2. 简述实验 → 3. 主要结果 → 4. 小结论

Methods:

- 足够别人复现
- 顺序与Results一致
- 不写结果

Introduction:

1. 背景与问题
2. 重要性：复述相关研究，提供 rationale
3. 假设/实验设计概述/关键结果预告

Discussion:

结构灵活，可以包括：总结核心发现，对比相关研究，讨论优势/局限，放入更大背景（理论/应用），给出展望

Abstract:

- 通常三件事：Rationale → Methods & Results → Conclusion
- 通常不用缩写，不放参考文献

Abstract范例一

Adenosine monophosphate (AMP)-activated protein kinase (AMPK) is a regulator of cellular catabolism that is activated by AMP. As AMP accumulates in cells with low ATP, AMPK is considered a stress-activated kinase. While studying organ growth during *Drosophila* development, we find that AMPK can also be activated by a signalling metabolite not related to stress. Specifically, we find that two physiological inputs known to regulate organ growth rates (ecdysone (a steroid hormone) and dietary protein) modulate expression of adenosine deaminase in the intestine. This, in turn, alters circulating adenosine levels. Circulating adenosine acts as a signalling molecule by entering cells, becoming phosphorylated to AMP and activating AMPK to inhibit organ growth. $\boxed{\text{Thus}}$, AMPK activity is regulated developmentally, and AMPK activity in one tissue can be remote controlled by another tissue via circulating adenosine. $\boxed{\text{Notably}}$, this mechanism accounts for half the effect of dietary protein on tissue growth rates in *Drosophila*.

Introduction

Methods & Results

Conclusion

Importance

Zhang, Y., Strassburger, K. & Teleman, A.A. Remote control of AMPK via extracellular adenosine controls tissue growth. *Nat Cell Biol* **27**, 1827–1837 (2025).

Abstract范例二

We lack tools to edit DNA sequences at scales necessary to study 99% of the human genome that is noncoding. To address this gap, we applied CRISPR prime editing to insert recombination handles into repetitive sequences, up to 1697 per cell line, which enables generating large-scale deletions, inversions, translocations, and circular DNA. Recombinase induction produced more than 100 stochastic megabase-sized rearrangements in each cell. We tracked these rearrangements over time to measure selection pressures, finding a preference for shorter variants that avoided essential genes. We characterized 29 clones with multiple rearrangements, finding an impact of deletions on expression of genes in the variant but not on nearby genes. This genome-scrambling strategy enables large deletions, sequence relocations, and the insertion of regulatory elements to explore genome dispensability and organization.

Gap

Methods & Results

Conclusion

Jonas Koeppel *et al.*,Randomizing the human genome by engineering recombination between repeat elements. *Science* **387**, eado3979 (2025).

Zotero使用教程参考
任意推送/视频

Abstract范例三

Studying the spatial distribution of cell types in tissues is essential for understanding their function in health and disease. A widely applied measure in spatial omics analysis is the pairwise neighbor preference of cell types, indicating whether two cell types frequently occur in close proximity to each other. While various neighbor preference methods exist, there is no clear guidance for selecting one over another. Here we present a comprehensive comparison of existing neighbor preference analysis methods. We systematically evaluate the method's underlying analytical steps and introduce COZI, a combination of analysis steps not previously described. We compare results from existing methods and COZI with respect to their ability to distinguish distinct tissue architectures and to recover neighbor preference directionality using two tissue simulations and two biological datasets. Overall, we delineate method-specific strengths and limitations and demonstrate that COZI provides sensitive and directional neighbor preference analysis for quantification of spatial data.

Introduction

Gap

Methods & Results

Conclusion

Schiller, C., Ibarra-Arellano, M.A., Bestak, K. *et al.* Comparison and optimization of cellular neighbor preference methods for quantitative tissue analysis. *Nat Commun* **17**, 3514 (2026).

1. 写作

Courtesy of Dr. Cindy Koerner

2. 答辩

Courtesy of Dr. Maiwen Caudron-Herger

3. 其他

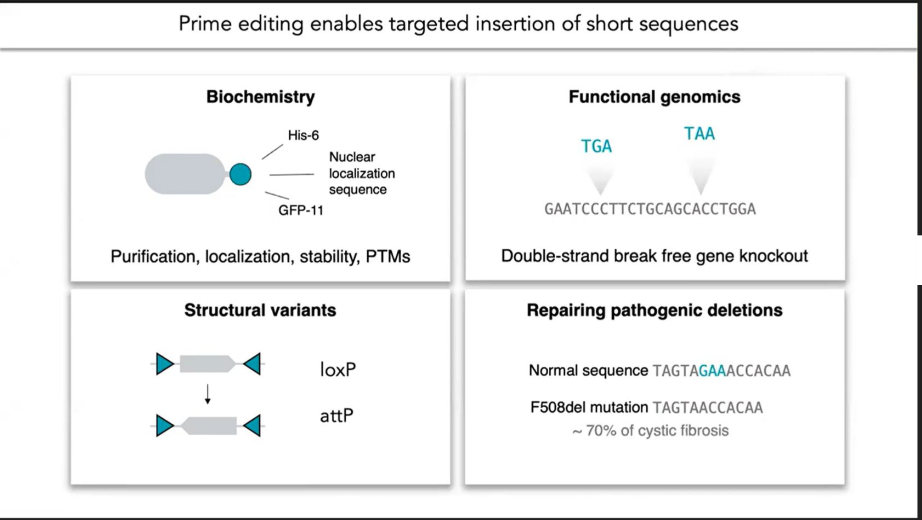
文档结构 → PPT结构

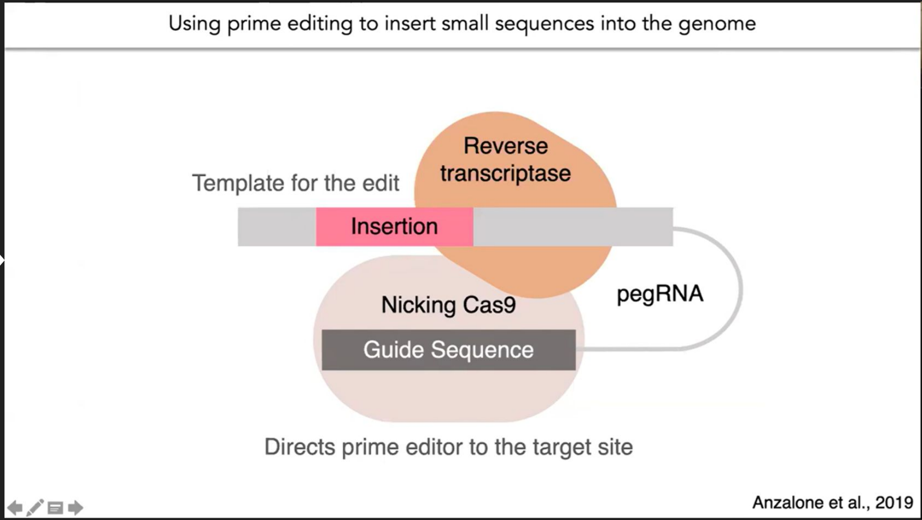
文档结构	在PPT里
Title	单独成页
Abstract	改造成大纲页
Introduction	适当压缩，**“从大到小”**
Methods	可以不单独成节，融入Results
Results & Figure	主体，**每页一个结论**
Discussion	适当压缩，**“从小到大”**（总结 → 展望）
References	建议逐页标注

文档结构 → PPT结构

文档结构	在PPT里
Title	单独成页
Abstract	改造成大纲页
Introduction	适当压缩，“从大到小”
Methods	可以不单独成节，融入Results
Results & Figure	主体，每页一个结论
Discussion	适当压缩，“从小到大”（总结 → 展望）
References	建议逐页标注

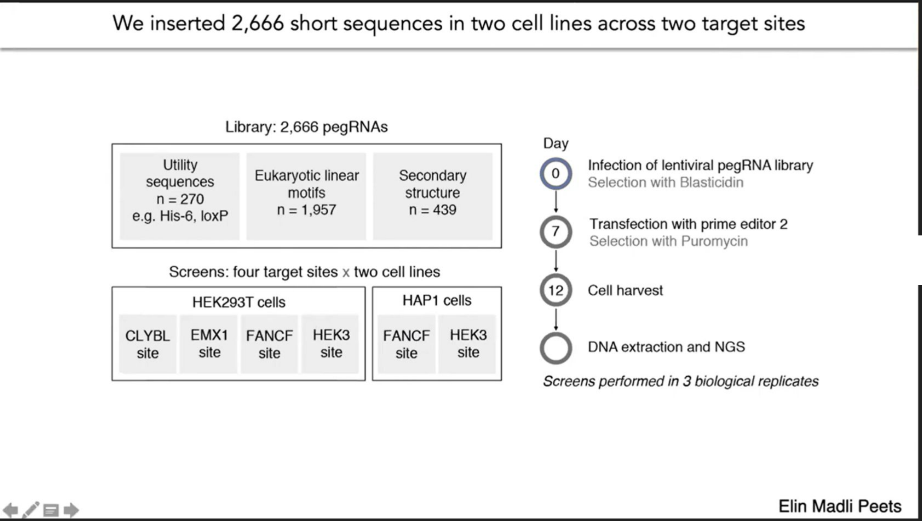
Introduction例子

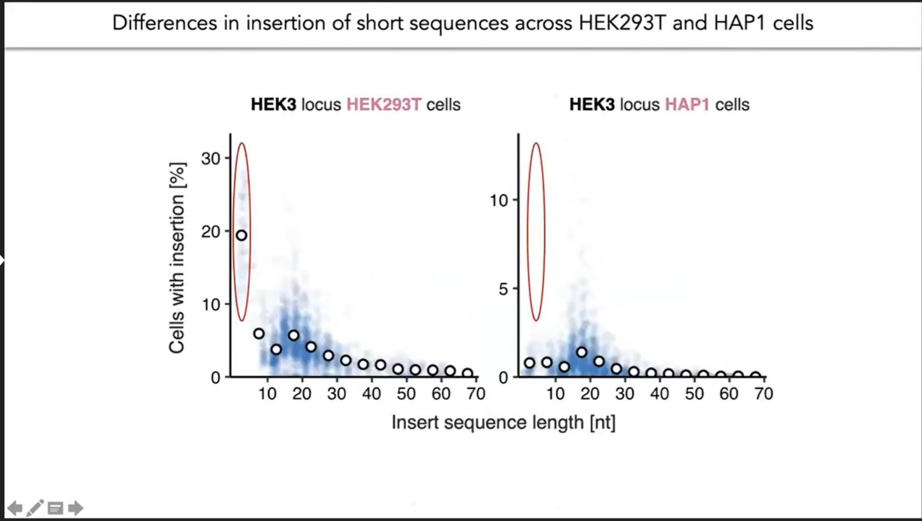




Koeppel, J., Weller, J., Peets, E.M. *et al.* Prediction of prime editing insertion efficiencies using sequence features and DNA repair determinants. *Nat Biotechnol* **41**, 1446–1456 (2023).

Methods & Results例子





Koeppel, J., Weller, J., Peets, E.M. *et al.* Prediction of prime editing insertion efficiencies using sequence features and DNA repair determinants. *Nat Biotechnol* **41**, 1446–1456 (2023).

Thinking, Fast and Slow

System 1 is your target!!!

- **Fast** and automatic
- Intuition
- Less mental effort
- Error prone…

System 2

- **Slow** and deliberate
- Concentration
- Mental effort
- Lazy…

“Thinking, Fast and Slow”, Daniel Kahneman

Take-home messages

- Adapt to your audience and aims
- Talk to System 1 (intuitive, effortless)
- Slide content: Less is more!
- Use alignment, animation, color, and font wisely
- Practice and get feedback
- Dress accordingly

1. 写作

Courtesy of Dr. Cindy Koerner

2. 答辩

Courtesy of Dr. Maiwen Caudron-Herger

3. 其他

- LLM的使用：
  - 避免“Ctrl C + V”
  - 只在deadline前帮助过关
  - “学而不用AI则罔，用AI而不学则殆”

- LLM的使用：
    - 避免“Ctrl C + V”
    - 只在deadline前帮助过关
    - “学而不用AI则罔，用AI而不学则殆”

- 探索校内资源（“星火”、“闯世界”、暑研/寒研等），找到自己的路！

All the best!

王一璐
yilu.wang@dkfz.de